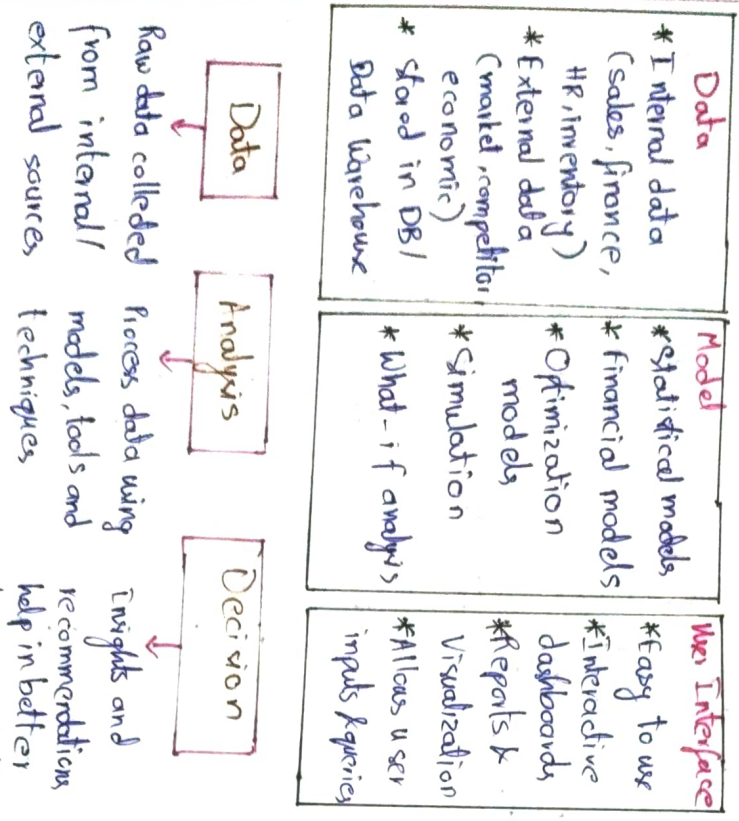


1. Decision Support Systems (DSS) :-

- * Computer-based systems that help managers in making semi-structured and unstructured decisions.
- * DSS supports decision makers by providing relevant information, models, and analysis tools.

Core Components of DSS :-



Benefits of DSS :-

- * Better decision quality
- * Cost reduction
- * Increased organizational performance
- * Competitive advantage

Types of DSS :-

- * Communication-Driven DSS
 - Supports communication and collaboration among decision makers.
- * Data-Driven DSS
 - Focuses on data retrieval, analysis and reporting.
- * Document-Driven DSS
 - Manages, retrieves and analyzes documents and information.
- * Model-Driven DSS
 - Uses models for analysis, forecasting and what-if scenarios.
- * Knowledge-Driven DSS
 - Uses knowledge bases and AI techniques for decision support.

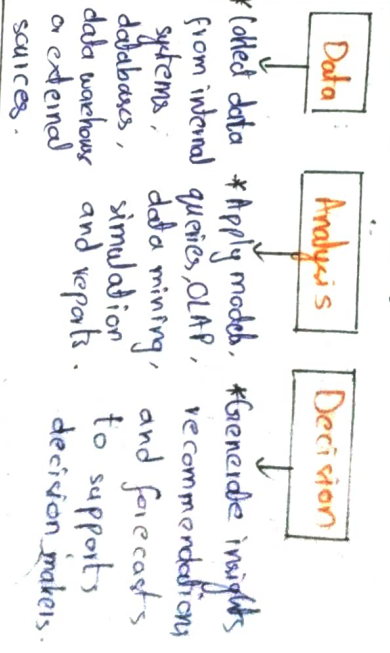
Real-World Example :-

Ex: Retail Sales DSS.
Outcome: Increase sales, reduced stockouts better customer targeting.

Role in Business Decision-Making :-

- * Improves quality of decisions.
- * Speeds up decision-making process.
- * Handles complex and unstructured problems.
- * Reduces risks and uncertainty.

How DSS Works? (Flow)



Applications of DSS :-

- * Marketing
- * Finance
- * Operations
- * HR
- * Supply Chain
- * Healthcare

Key Characteristics of DSS :-

- * Interactive and flexible
- * Supports semi-structured decisions
- * User-controlled
- * Integrates data and models
- * Enhances human decision-making

Operational Systems (OLTP) vs Decision Support Systems (DSS/OLAP)

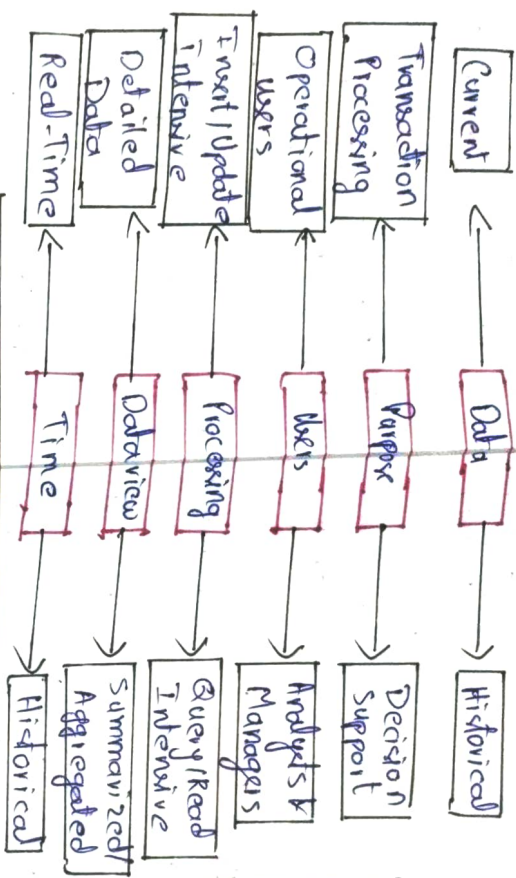
Operational Systems (OLTP)

Handle day-to-day transactions and operations.

Key Characteristics

Data Type	Current, Detailed, Transaction-oriented
Purpose	Run the business (record & process transactions)
Users	Clerks, staff, Operational users
Processing	Frequent inserts/updates, High volume, Simple
Data	Real-time / Current
Examples	Banking transactions, Order processing, Inventory, Billing

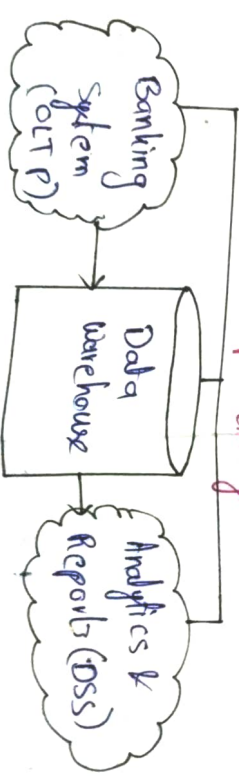
Key Differences



Similarities & Interaction

Both use a common data source. OLTP feeds data to Data warehouse. Together support efficient business operations and better decisions

Example linkage

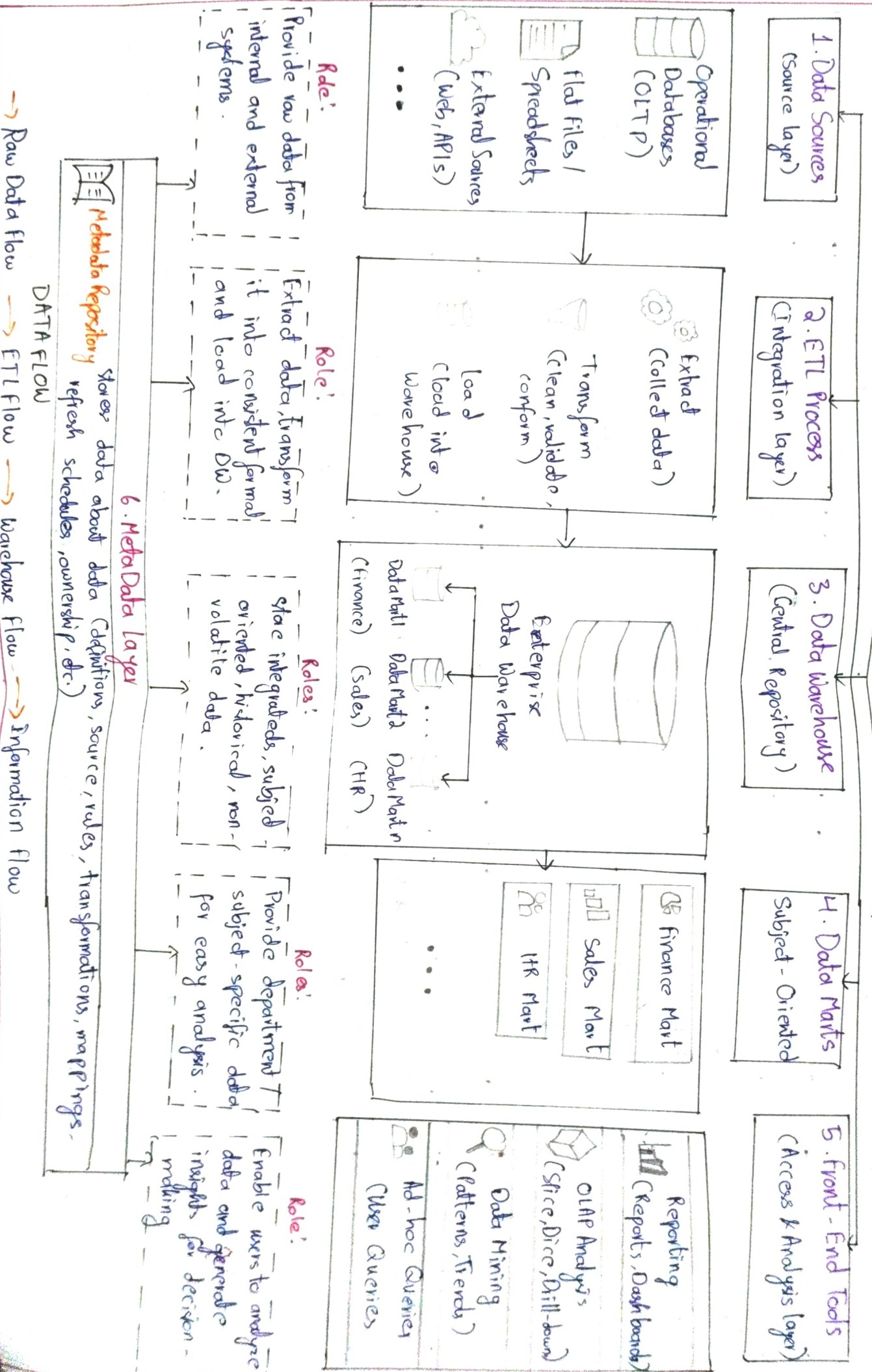


Decision Support Systems (DSS/OLAP)

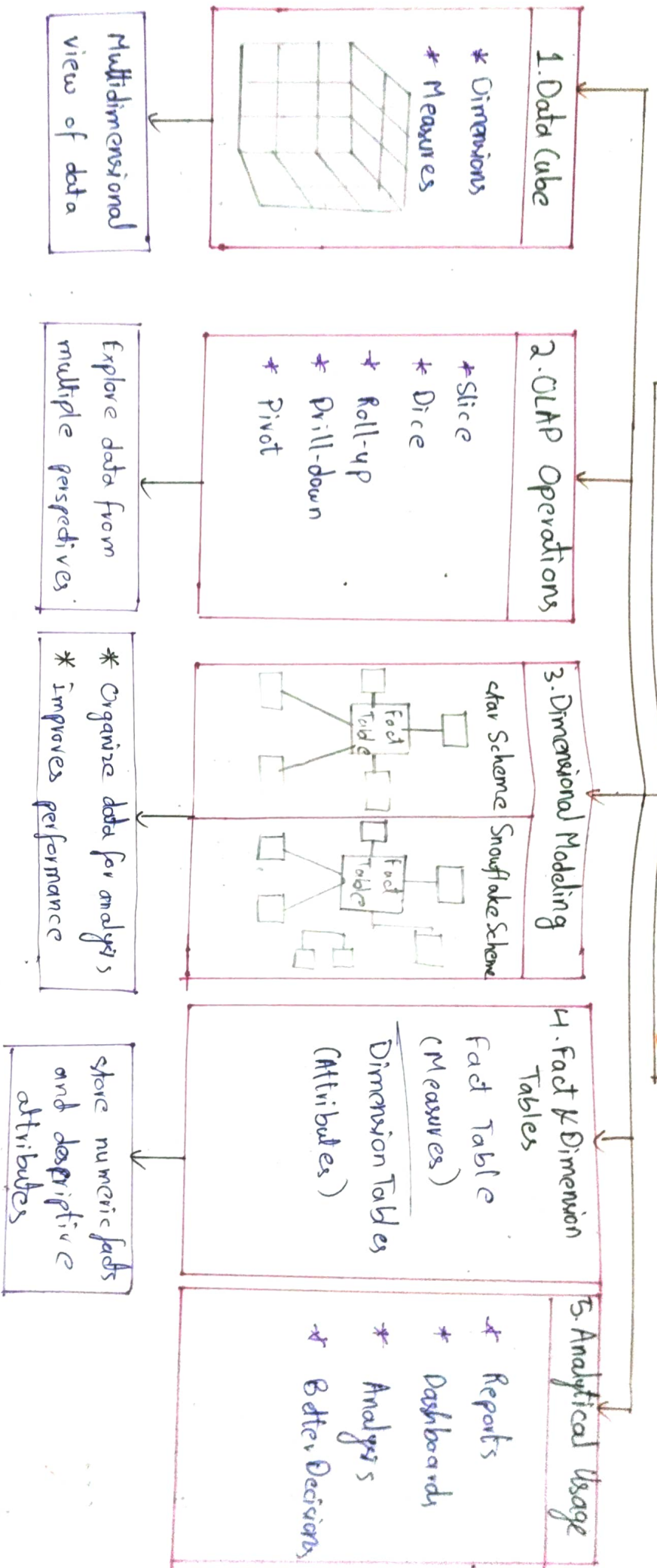
Key Characteristics

Data Type	Historical, Summarized, Subject-oriented
Purpose	Support decision-making, Analysis & Planning
Users	Managers, Analysts, Executives
Processing	Complex queries, Aggregations, Read-heavy
Data	Historical / Time-variant
Examples	Sales analysis, Forecasting, Trend analysis, Dashboards, Reports

Architecture of Data warehouse



OLAP, Data Cubes & Dimensional Modeling



DATA



OLAP QUERY



DATA CUBE



INSIGHTS

Dimensional Modeling → Builds Data Cube → enables OLAP Operations → Provides Insights

Query Processing & BI Tools (KNIME)

